

Crystal Characterization for Dual Readout Calorimetry

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Abstract

High-resolution calorimetry for future collider experiments can be achieved by employing dual readout (DR) of Cherenkov and scintillation light. The CalVision consortium aims to realize this state-of-the-art calorimetric measurement for both homogeneous scintillating crystals and fiber-based hadronic sections [1], with a phased program that includes investigating candidate crystal materials for the electromagnetic calorimeter. This work focuses on characterizing the optical properties of Bismuth Germanate (BGO) samples as a potential crystal candidate by measuring its optical transmission spectrum, spatial uniformity of light yield, scintillation decay time, and X-ray-excited emission spectrum to assess its suitability for precise separation of Cherenkov and scintillation signals and to guide material selection for next-generation dual-readout calorimeters.

1 Introduction and Motivation

Precision measurements allow experimenters to probe physics at very high energy scales via loop corrections to Standard Model (SM) processes, detect minute deviations from SM predictions indicative of new physics, discriminate rare signals from copious backgrounds at high-rate experiments, and achieve sensitivity to the very weak interactions of candidate dark matter particles. Precision measurements of the visible energy in calorimeters combined with tracking information are crucial for the reconstruction of jets, vector and Higgs bosons, top quarks, overall event topologies, recoil masses, and for missing energy measurements that are key to many beyond-SM physics searches.

The signal produced by a calorimeter shower initiated by an incident hadron is generally lower than that of an electron/photon of the same energy, due to invisible energy losses from, e.g., nuclear binding energies, neutrinos, and particles with small inelastic cross sections such as neutrons escaping the detector or depositing energy outside of the sampling time window. The amount of missing energy is correlated with the number of inelastic nuclear collisions. Maximal-information calorimetry identifies measurable quantities correlated with the missing energy to enable shower-by-shower corrections. An example is dual-readout calorimetry, which uses scintillation light to measure the total ionizing energy deposit, and timing and/or wavelength filters to isolate and collect the Cherenkov light, which is produced only by relativistic particles in the shower (mainly electrons produced in pion showers). As shown in Figure 1, the two types of light can be separated either by their wavelength (left), their arrival time (right), or polarization (not shown). As illustrated in the figure, the scintillation spectrum for $PbWO_4$ begins very close to where the transmission cuts off for small wavelengths, and the optimal region to select a Cherenkov signal is then above $\simeq 600$ nm. For scintillators such as BGO, where the scintillation spectrum is further shifted from the transmission cutoff, a notch filter may be used to select light on both sides of the scintillation spectrum [2].

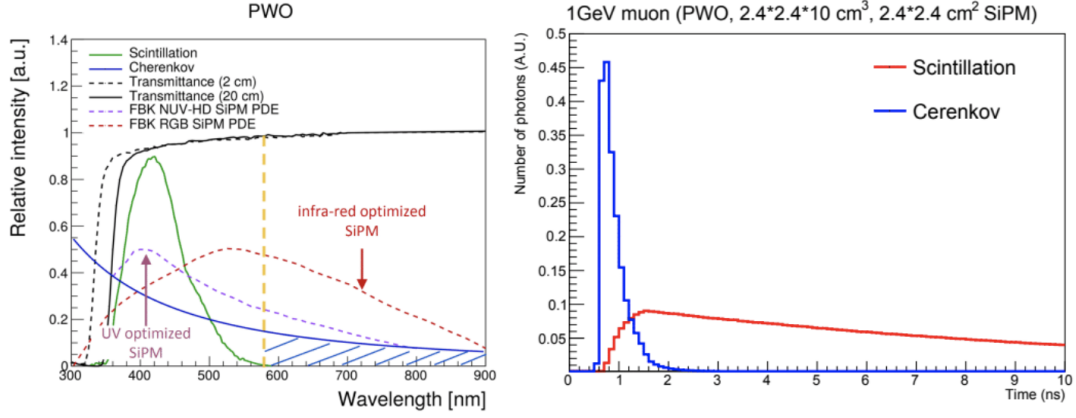


Figure 1: [left] Wavelength spectra of Cherenkov and scintillation emissions of $PbWO_4$ crystals along with the transmittance curves and photon detection efficiencies of example SiPMs. Hashed areas represent potential Cherenkov detection regions using wavelength filters. [right] The timing distributions of scintillation and Cherenkov light at the end of a 100 mm long $PbWO_4$ crystal with a cross-sectional area of $24 \times 24 \text{ mm}^2$, produced by 1 GeV muons crossing the middle of the crystal bar.

This work will focus on studying the optical quality of the BGO samples, which helps characterize the crystals based on properties such as scintillation signal region, transmission spectrum, decay time period, and net light-yield spectrum. I plan to study two BGO samples with dimensions $22 \times 22 \times 160 \text{ mm}^3$ using a state-of-the-art spectrophotometer, PMTs, and Data Acquisition Systems (DAQ) [3]. The proposed timeline is:

- **Week 1:** Procuring and measuring physical dimensions of the crystals; measuring X-ray Excited Luminescence. (DONE)
- **Week 2:** Light Output and Decay Time measurements for the scintillation signal; measurements of the light-response uniformity of the crystal. (DONE)
- **Week 3:** Measurement of the transmission spectrum of the crystal.
- **Week 4:** Analyzing collected data and presenting the first draft of the report.
- **Week 5:** Submission of final draft and presentation of the project.

References

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- [3] Liyuan Zhang and Ren-Yuan Zhu. Progress of inorganic scintillators for future hep experiments, 2024.